

PLC Programming & Electro-Pneumatic Sequence Control

Ladder logic, digital I/O, and automated two-cylinder sequencing with an Allen-Bradley PLC

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Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
Team	Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortez, Abhin Ghandi, Ramid Brown	Equipment	Amatrol PLC trainer with I/O simulator switches, RSLogix 500, electro-pneumatic application panel, compressed air, process interface unit

PROJECT SUMMARY

This lab explored programmable logic controllers (PLCs) in industrial automation. Using Rockwell’s RSLogix 500 software and a PLC trainer, the team identified the system’s digital inputs and outputs and observed how normally open (NO) and normally closed (NC) contacts in ladder logic control output behavior. The PLC was then connected to an electro-pneumatic application panel, where a ladder program used solenoid valves and limit-switch feedback to run a fully automatic two-cylinder extend-and-retract sequence without any operator input after start-up.

1. Introduction

Industrial automation depends on control systems that handle complex tasks precisely and repeatably. PLCs let engineers automate machinery with programmed logic instead of hardwired relay circuits, making systems easier to modify and troubleshoot. This lab covered PLC operation, ladder logic programming, and electro-pneumatic sequence control.

2. System Components

Table 1. Lab hardware and software

Component	Role
PLC with I/O simulator switches	Executes the ladder program; switches simulate sensor inputs and lamps show outputs, allowing logic to be tested safely
RSLogix 500	Used to write, download, monitor, and troubleshoot ladder logic
Electro-pneumatic application panel	Solenoid valves, limit switches, and two pneumatic cylinders driven by PLC outputs
25-pin interface cable	Links the PLC trainer’s I/O to the application panel
Compressed air supply	Powers the cylinders so output signals produce physical motion
Process interface unit / basic process rig	Signal link and test platform for process control demonstrations



Figure 1. PLC trainer and application panel

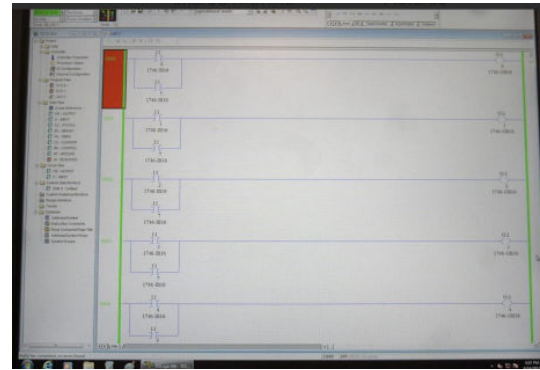


Figure 2. Ladder program in RSLogix 500

3. Procedure

The computer was connected to the PLC and RSLogix 500 was started. The processor file *iotest* was loaded to study how NO and NC contacts behave, using combinations of simulator switches and watching the indicator lamps; the program *test1* was used to map each input and output to its slot and terminal address. The trainer was then connected to the electro-pneumatic panel through the 25-pin cable, compressed air was turned on, and the processor file *test4* was loaded to run the automatic cylinder sequence while the solenoid valves and limit switches were observed.

4. Results

Table 2. I/O map for the sequence program

Device	Address	Function	Rung
Manual switch	I:1/0	Starts the cycle	0000
Limit switch 1LS	I:1/2	Top cylinder at home	0003
Limit switch 2LS	I:1/3	Top cylinder fully extended	0002
Limit switch 4LS	I:1/4	No effect on the logic	—
Limit switch 5LS	I:1/6	Bottom cylinder fully extended	0001
Solenoid 2A	O:2/1	Extend bottom cylinder	0000
Solenoid 1A	O:2/0	Extend top cylinder	0001
Solenoid 2B	O:2/2	Retract bottom cylinder	0003
Internal bit	B3:0/0	Latch that ends the top cylinder stroke	0002

Table 3. Automatic sequence (4LS toggled during the run had no effect)

Step	Event	PLC response	Motion
1	Manual switch turned on (I:1/0)	Rung 0000 energizes and latches solenoid 2A (O:2/1)	Bottom cylinder extends
2	Bottom cylinder reaches 5LS (I:1/6)	Rung 0001 energizes solenoid 1A (O:2/0)	Top cylinder extends
3	Top cylinder reaches 2LS (I:1/3)	Rung 0002 sets latch B3:0/0, which opens rung 0001 and drops solenoid 1A	Top cylinder returns home
4	Top cylinder reaches 1LS (I:1/2)	Rung 0003 energizes solenoid 2B (O:2/2), breaking the rung 0000 latch	Bottom cylinder retracts
5	Both cylinders home	Latches reset	System back to its initial state

5. Discussion

Connecting RSLogix 500 to the trainer made it possible to watch the controller scan physical inputs and convert them into outputs in real time. Testing NO and NC contacts showed how the same input can either enable or block a rung depending on how it is programmed, which is the foundation of ladder logic design.

The electro-pneumatic sequence was the clearest demonstration of feedback-driven control. Each step began only after a limit switch confirmed that the previous motion had finished: the bottom cylinder extended, the top cylinder extended and returned, and then the bottom cylinder retracted to complete the cycle. The internal latch bit B3:0/0 and the NC contacts prevented conflicting outputs, such as extending and retracting the same cylinder at once. Because the logic depended on confirmed positions rather than timing, the sequence ran precisely and repeatably with no operator involvement.

6. Conclusion

This lab showed why PLCs are central to industrial automation. Structured ladder logic turned digital inputs into controlled physical outputs, and the combination of solenoid valves and limit-switch feedback produced a coordinated, repeatable multi-stage cylinder sequence. The exercise also highlighted that reliable automation depends on integrating mechanical hardware, electrical signals, and well-designed logic, a foundation for understanding the larger process control systems used in industry.

References

- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.